

Day 4

Nina White

July 17, 2009

1 A Crash Course in Complex Numbers

The complex numbers are defined to be

$$\{z = a + bi \mid a, b \in \mathbb{R}\}$$

The number a is called the *real part* of z , denoted $\Re z$, and b is called the *imaginary part* of z , denoted $\Im z$. (You can just use “Re” and “Im”.) We plot complex numbers on the plane, using coordinates $(\Re z, \Im z)$. We add and multiply like this:

$$(a + bi) + (c + di) = (a + c) + (b + d)i \text{ (like vector addition)}$$

$$(a + bi)(c + di) = ac - db + (bc + ad)i$$

The second formula can be derived with usual distribution if we let $i^2 = -1$. Note that i is not the *only* square root of -1 . Can you name another?

If $z = a + bi$, we define the *complex conjugate* of z to be

$$\bar{z} := a - bi$$

Since a complex number is geometrically just a point in the plane, it can be specified by its polar coordinates, (r, θ) in the plane. The *norm* of $z = a + bi \in \mathbb{C}$ is its distance from the origin, so, using the Pythagorean theorem:

$$|z| = \sqrt{a^2 + b^2}$$

The *argument* of z , denoted $\arg z$, is the angle the vector (a, b) makes with the positive x -axis. This is only defined modulo 2π . With this new notation, the polar coordinates of $z \in \mathbb{C}$ are $(|z|, \arg z)$. While complex addition is simply vector addition, there is a nice formula for vector multiplication in terms of polar coordinates:

$$(r, \theta)(r', \theta') = (rr', \theta + \theta') \tag{1}$$

That is, we multiply the norms and add the arguments. This can be proved using a couple of lemmas. We know

$$z = |z| \frac{z}{|z|},$$

so z is a product of its norm times a complex number of unit norm ($\frac{z}{|z|}$ has unit norm by Exercise 1 (f)). In other words, it is expressible as

$$|z|(\cos \theta + i \sin \theta) =: |z| \operatorname{cis} \theta,$$

where $\theta = \arg z$. Once we have that, Equation 1 is a calculation using the sin and cos sum of two angles formulas and can be rewritten

$$(r_1 \operatorname{cis} \theta_1)(r_2 \operatorname{cis} \theta_2) = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2) \quad (2)$$

Moral: multiplication is a composition of a rotation and dilation about the origin. See Exercise ?? for more on this.

2 Möbius Transformations

A *Möbius transformation* (or *Linear Fractional Transformation*, *LFT*) is a non-constant map ϕ of the form

$$\phi(z) = \frac{az + b}{cz + d}$$

where $a, b, c, d \in \mathbb{C}$.

The *Riemann sphere* (or *extended complex plane*) $\overline{\mathbb{C}}$, is formally defined to be $\mathbb{C} \cup \{\infty\}$. But it's more than just a set, it's homeomorphic to a sphere. We'll discuss that more tomorrow using stereographic projection. For now it's enough to think of ∞ as a formal element we use in our Möbius transformations. For example, we'll use the following rules when calculating $\phi(w)$ for $w \in \overline{\mathbb{C}}$:

- If $cw + d = 0$, $\phi(w) = \infty$. (Note that we won't simultaneously have $aw + b = 0$, otherwise our function would be constant.)
- If $a, c \neq 0$, $\frac{a\infty+b}{c\infty+d} = \frac{a}{c}$.
- If $c = 0$ (hence $a \neq 0$ to avoid ϕ being constant), then $\phi(\infty) = \infty$.
- If $a = 0$ (hence $c \neq 0$ to avoid ϕ being constant), then $\phi(\infty) = 0$.

These rules are summarized as: $\frac{1}{0} = \infty$, $\frac{\infty}{1} = \infty$, $\frac{\infty}{\infty} = 1$. Note that we'll never get $\frac{\infty}{0}$ or $\frac{0}{\infty}$.

3 Exercises

Exercise 1 (L). Show the following useful identities:

- (a) $\Re z = \frac{1}{2}(z + \bar{z})$
- (b) $\Im z = \frac{1}{2i}(z - \bar{z})$
- (c) $z\bar{z} = |z|^2$
- (d) $\overline{a+b} = \overline{a} + \overline{b}$

- (e) $\overline{ab} = \overline{a} \cdot \overline{b}$
- (f) $|zw| = |z||w|$
- (g) Verify the multiplication formula, Equation 2.
- (h) Find $1/\sqrt{2+5i}$. Hint: use (c).

Exercise 2 (L). Show that any Möbius transformation can be obtained as a composition of the following maps.

$$\begin{aligned} z &\longrightarrow az \\ z &\longrightarrow z + b \\ z &\longrightarrow \frac{1}{z} \end{aligned}$$

Exercise 3 (L). Check that on the domain $\overline{\mathbb{C}}$, any Möbius transformation is a bijection. Check that the set of Möbius transformations forms a group.

Exercise 4 (L). Show using algebra that any Möbius transformation not equal to the identity has one or two fixed points. (Hint: consider first the case that $c = 0$, then then the case $c \neq 0$.)

*Exercise 5 (L, verbatim from Donald Sarason's *Notes on Complex Function Theory*).* If z_1, z_2, z_3 are three distinct points of $\overline{\mathbb{C}}$, and w_1, w_2, w_3 are three distinct points of $\overline{\mathbb{C}}$, then there is a unique Möbius transformation ϕ such that $\phi(z_j) = w_j$, $j = 1, 2, 3$. We'll prove this in steps.

- (a) First show uniqueness using Exercise 4.
- (b) Let $w_1 = \infty, w_2 = 0, w_3 = 1$. Find ϕ if
 - z_1, z_2, z_3 are all finite
 - $z_1 = \infty$
 - $z_2 = \infty$
 - $z_3 = \infty$
- (c) Using Exercise 3, how does this show existence in our theorem?
- (d) Contrast this theorem to 3-point determinacy of $\text{Isom } \mathbb{E}^2$.

Exercise 6 (C). Try to describe what the map $z \longrightarrow \frac{1}{z}$ does to the plane.

Try to describe what the map $z \longrightarrow \frac{1}{\bar{z}}$ does to the plane. (Hint: you'll need the technique from Exercise 1 (h)).

Exercise 7 (from Sarason). Exhibit the Möbius transformation that maps $0, 1, \infty$ to $1, \infty, -i$ respectively.

Exercise 8 (E). Check that matrices of the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, $a, b \in \mathbb{R}$, form a subgroup of $GL(2, \mathbb{R})$. Show that this group is isomorphic to $\mathbb{C} - \{0\}$.